THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 10: **Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $a : b = 4 : 9$ and $b : c = 6 : 5$ and $a + b + c = 369$, then b equals: 1

(a) 162 **(b)** 171 **(c)** 135 **(d)** 72

2. 90 men have provisions for 24 days. After 6 days 30 men leave. The remaining provisions last the rest of the men (days): 1

(a) 29 **(b)** 36 **(c)** 18 **(d)** 27

3. A vessel has 45 L milk and 20 L water. The litres of water to be added to make the ratio of milk to water 3:2 is: 1

(a) 25 **(b)** 10 **(c)** 20 **(d)** 13

4. The wages of 8 workers for 25 days are ₹ 36000. The wages of 15 workers for 12 days at the same daily rate (in ₹) are: 1

(a) 67500 **(b)** 32400 **(c)** 36000 **(d)** 32580

5. The present ages of Ria and Sia are in the ratio 4:7. After 10 years the ratio will be 2:3. Ria's present age (years) is: 1

(a) 30 **(b)** 18 **(c)** 35 **(d)** 20

6. Pipe A fills a tank in 20 hours and pipe B in 5 hours. Together they fill it in (hours): 1

(a) 25 **(b)** 5 **(c)** 4 **(d)** 12.5

7. In a pie chart a 54° sector represents 21 students. The total number of students is: 1

(a) 70 **(b)** 756 **(c)** 161 **(d)** 140

8. If $a : b = 5 : 7$ and $b : c = 14 : 9$ and $a + b + c = 264$, then b equals: 1

(a) 120 **(b)** 72 **(c)** 112 **(d)** 80

9. 80 men have provisions for 36 days. After 12 days 20 men leave. The remaining provisions last the rest of the men (days): 1

(a) 32 **(b)** 24 **(c)** 48 **(d)** 34

10. A vessel has 28 L milk and 8 L water. The litres of water to be added to make the ratio of milk to water 4:3 is: 1

(a) 13 **(b)** 20 **(c)** 16 **(d)** 26

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A tank is filled by pipe A in 36 hours and by pipe B in 12 hours. How long will both take together? 2

12. The present ages of a father and son are in the ratio 8:3. After 10 years the ratio will be 2:1. Find their present ages. 2

13. If $x : y = 7 : 3$, find the ratio $(2x + 5y) : (3x - 4y)$. 2

14. A tank is filled by pipe A in 40 hours and by pipe B in 10 hours. How long will both take together? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A garrison of 300 men has provisions for 30 days. After 6 days 100 men leave. For how many more days will the remaining provisions last for the men who stay? 3

16. A mixture contains milk and water in the ratio 4:5. When 8 L of water is added, the ratio of milk to water becomes 4:7. Find the quantity of milk and water in the original mixture. 3

17. In a pie chart, the sector for Science is 24° and represents 8 students, while the sector for Maths is 60°. Find the total number of students and the number who like Maths. 3

OR

A garrison of 160 men has provisions for 50 days. After 20 days 40 men leave. For how many more days will the remaining provisions last for the men who stay? 3

18. A mixture contains milk and water in the ratio 7:3. When 15 L of water is added, the ratio of milk to water becomes 7:8. Find the quantity of milk and water in the original mixture. 3

OR

In a pie chart, the sector for Science is 36° and represents 13 students, while the sector for Maths is 108°. Find the total number of students and the number who like Maths. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The present ages of Anil and Bina are in the ratio 4:9. After 8 years the ratio of their ages will be 6:11. Find their present ages and the ratio of their ages 8 years ago. 5

OR

20 workers can finish a job in 36 days. After 12 days 5 workers leave. In how many total days will the job be finished? By how many days is it delayed? 5

20. The present ages of Anil and Bina are in the ratio 2:5. After 6 years the ratio of their ages will be 5:11. Find their present ages and the ratio of their ages 6 years ago. 5

OR

36 workers can finish a job in 20 days. After 5 days 9 workers leave. In how many total days will the job be finished? By how many days is it delayed? 5

END OF QUESTION PAPER — Total: 40 Marks